NO.8 INEFFECTIVE OPERATION OF A/C DURING ENGINE STOP (i-stop system) [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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8	INEFFECTIVE OPERATION OF A/C DURING ENGINE STOP (i-stop SYSTEM)
DESCRIPTION	• The A/C airflow temperature is too high or low compared with the set temperature while the i-stop function is operating.
POSSIBLE CAUSE	**A/C system malfunction** • Climate control unit falsely recognizes MAX COLD of air mix door on driver-side — Driver-side air mix actuator malfunction — Driver-side air mix actuator position sensor malfunction — Driver-side air mix door link stuck

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**DETERMINE IF MALFUNCTION CAUSE IS i-stop SYSTEM OR A/C SYSTEM** • Verify the malfunction symptom. • Does the malfunction occur only while the i-stop function is operating (engine stopped)?	Yes	Go to the next step.
		No	Go to the applicable A/C malfunction diagnostic procedure. (See TROUBLESHOOTING INDEX [FULL-AUTO AIR CONDITIONER].)
2	**VERIFY DTC** • Retrieve the PCM and climate control unit DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [FULL-AUTO AIR CONDITIONER].)
		No	Go to the next step.
3	**DETERMINE IF MALFUNCTION CAUSE IS DRIVER-SIDE AIR MIX ACTUATOR SIGNAL OR OTHER** • Measure the voltage at the following terminal (wiring harness-side) when the temperature setting is MAX COLD. — Climate control unit terminal G • Is the voltage normal? (See CLIMATE CONTROL UNIT INSPECTION [FULL-AUTO AIR CONDITIONER].)	Yes	The system is normal. • Effect is due to a possible change in the vehicle environment while the engine is stopped (change in solar radiation).
		No	Go to the next step.
4	**INSPECT DRIVER-SIDE AIR MIX ACTUATOR** • Inspect the driver-side air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is there any malfunction?	Yes	Replace the driver-side air mix actuator. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	Inspect the air mix actuator and linkage for sticking. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results.
5	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)		